Recovering 'lost' files

Yes, 'recovering lost files' means recovering the files that you've deleted 'forever' using *'Shift+Del'* or *'Empty Trash'!* Even files from formatted partitions can be recovered! You might be wondering how this works and how the disk could store all this stuff even though its capacity is limited. To figure that out, you should learn what happens when you delete a file. You might have noticed that deletion is faster than file copying. Also, the time required to delete a small file and a large file is the same. Hence, we understand that the OS doesn't erase every byte of a file to delete it. It simply removes the link to the file — that is why the process of deletion is known as 'unlinking.' The file remains there until new files need its space to reside. Only then does the file disappear from the hard drive. Even formatted disks allow us to recover deleted files. But it is always recommended to unmount the partition and perform the recovery immediately, once you find you've deleted an important file accidentally.

PhotoRec, Foremost and Scalpel are the most popular recovery utilities used in GNU/Linux. Here, we have chosen PhotoRec.

PhotoRec

PhotoRec is a very powerful tool that can recover files from a hard disk, digital camera and optical disks. It doesn't matter what the file system of the medium is, which means the files can be recovered even if the file system is unrecognised or if it has been severely damaged or formatted. PhotoRec won't write anything to the medium — that makes it a safe tool. It has been written by Christophe Grenier *<grenier@cgsecurity.org>*.

Note that PhotoRec comes with the package 'testdisk'. Your OS should already have it.

Before following the necessary steps, ensure that you have a good storage medium to collect the recovered files ('recovery medium'). It would be nice to use an external hard disk.

Now, let's see what the magical steps are:

0) Boot from a recovery disk (i.e., a live CD) if you are unable to load the native OS.
1) Unmount the partition from which the files are to be recovered, if possible.
2) Insert the recovery medium.
3) Open a terminal and give the following command:

```
sudo photorec
```

Follow the instructions to select the partition to be recovered and the recovery medium. 'Intel' would be the partitioning type in most cases — if you are asked for it. There is an option 'File Opt' under the disk selection dialogue box. Make use of it if you want to choose the file types.

4) You might have to change the owner of the recovered files since they are written under *'sudo'*. Run the command:

```
sudo chown <your-username> files Example:

sudo chown nandakumar /media/myUSBDevice/recup*
sudo chown nandakumar /media/myUSBDevice/recup*/*
```

PhotoRec stores files in the recovery medium under the folders titled *recup_dir.1*, *recup_dir.2*, etc. *recup** is used to address all these folders. *recup*/** means 'all files under all recup dirs'. In the above example, */media/myUSBDevice* is the medium used to store the recoverd files.

Recovering the entire file system

One day, during my usual GNU/Linux hacking, I accidentally wrote data to */dev/sda* directly (yes, I was using *sudo*). At the next start-up, the BIOS informed me that there was no hard disk found in my system! I realised with a shock that I was changing some of the first (important) bytes of my hard disk by writing data directly to */dev/sda*, making it unusable! My OS, my HDD and my valuable data — oh, no! But thanks to TestDisk, I recovered my entire hard disk with each byte of it unchanged.

What actually happens in similar cases is that only the 'partition table' of the hard disk becomes damaged. The files, folders and partitions are safe. This is almost similar to the situation we discussed in the earlier section. The cause is simple, but the consequences are terrible. You can't even format that disk! But the tool TestDisk comes to our help.

TestDisk

TestDisk is a partition recovery tool that can be used to recover lost partitions and make non-booting disks bootable again. It is also from the author of PhotoRec. Let's learn how to recover partitions using TestDisk.

1) Take a back-up of your important data if you can still access your hard disk.
2) Boot from a recovery disk and open a terminal. If you can still access your operating system, you can open the terminal from it. Run the command 'sudo testdisk' and follow the instructions. Make sure that you select the correct disks and partitions (*/dev/sda* will be the hard disk, normally). 'Search' is to search the lost partitions, and 'Write to disk' is to store the recovered partitions.

File system patchwork

Automatic file system check-ups are performed during the GNU/Linux boot process and the user is allowed to correct the problems with a single keystroke (e.g., 'Press F to fix, I to ignore and S to skip' — you don't have to ignore such situations). But sometimes the problems become severe and the kernel informs you it is unable to load. Most people would re-install the entire system when this error message appears. They don't know that this re-installation, which causes data loss and time loss, could be avoided by running